Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI

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Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

Review Rating column

$59.76

Avg Purchase

Demographics, purchase details, and shopping behavior — loaded into PostgreSQL for analysis.

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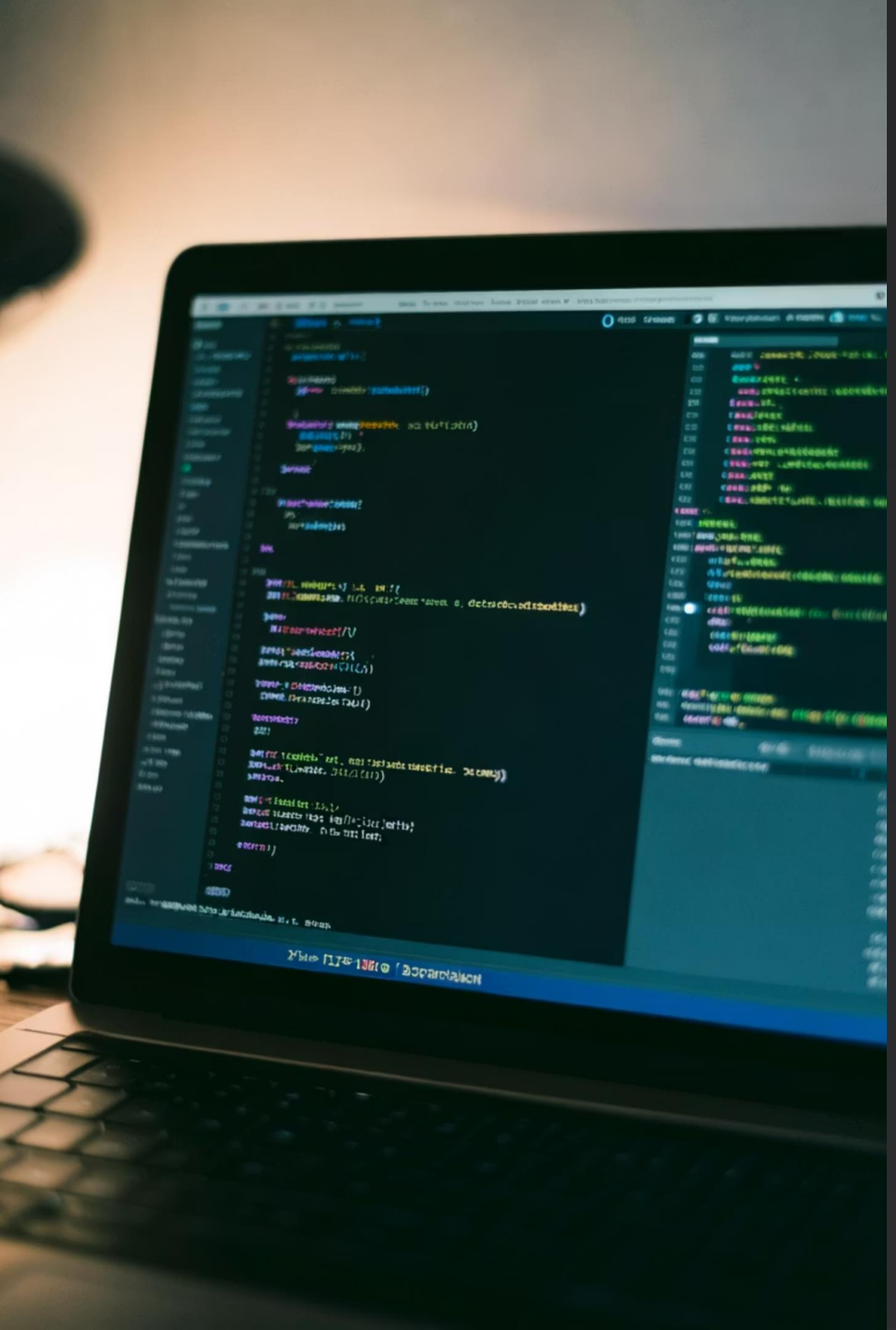

Python: Data Preparation

01

Load & Explore

pandas, df.info(), .describe()

02

Clean

Median imputation for Review Rating

03

Engineer

age_group, purchase_frequency_days

04

Load to DB

PostgreSQL integration

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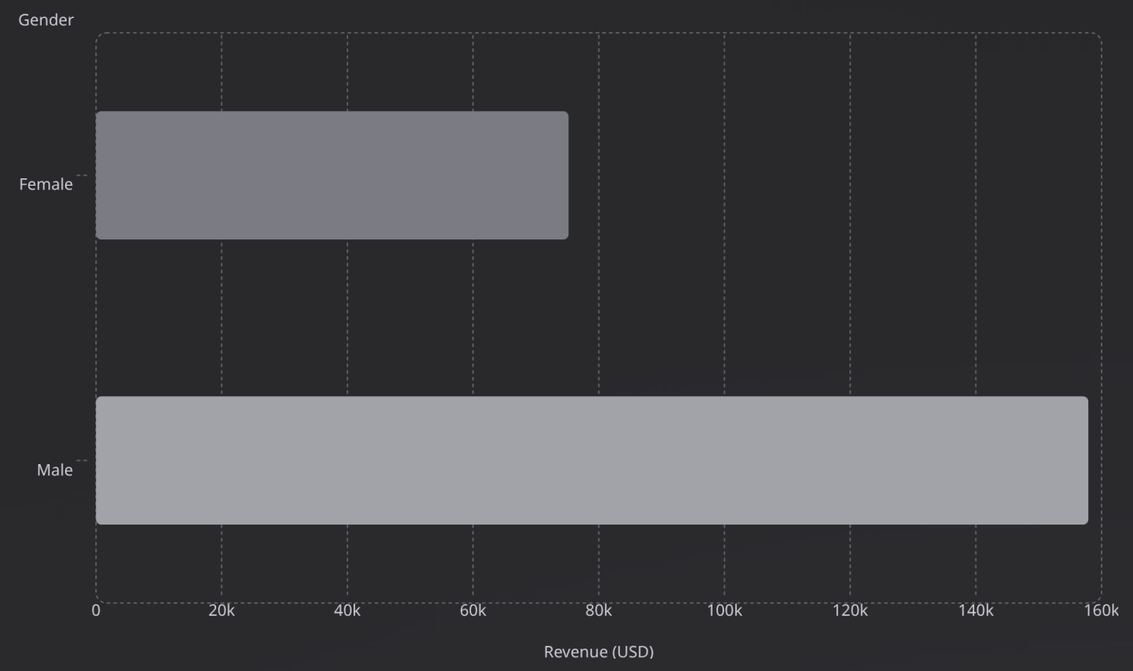
Revenue by Gender



Key Finding

Male customers generated **2.1× more revenue** than female customers — $157,890 vs $75,191.

Male purchases: 2,652 · Female purchases: 1,248

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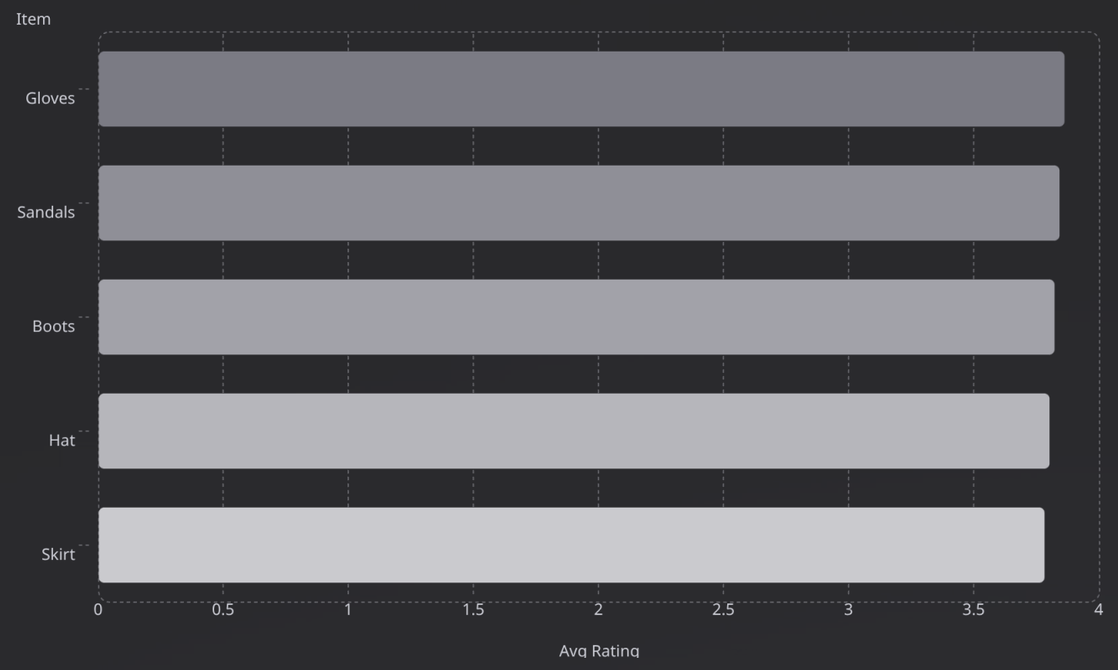
Top Products & Shipping Insights

Top 5 Products by Rating



Shipping Type — Avg Spend

Standard	Express
$58.46	$60.48

Express shippers spend slightly more per order.

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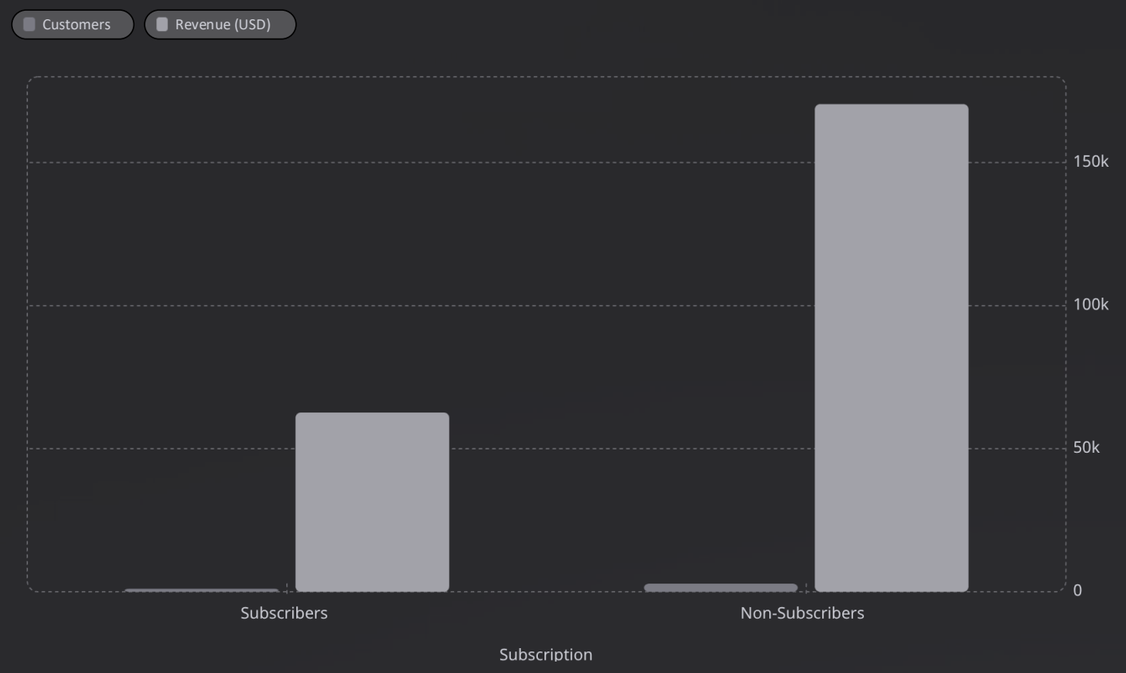
Subscribers vs. Non-Subscribers



Insight

Non-subscribers drive **73% of total revenue**, yet avg spend is nearly identical (~$59.50).

Only 27% of customers are subscribers — a major growth opportunity.

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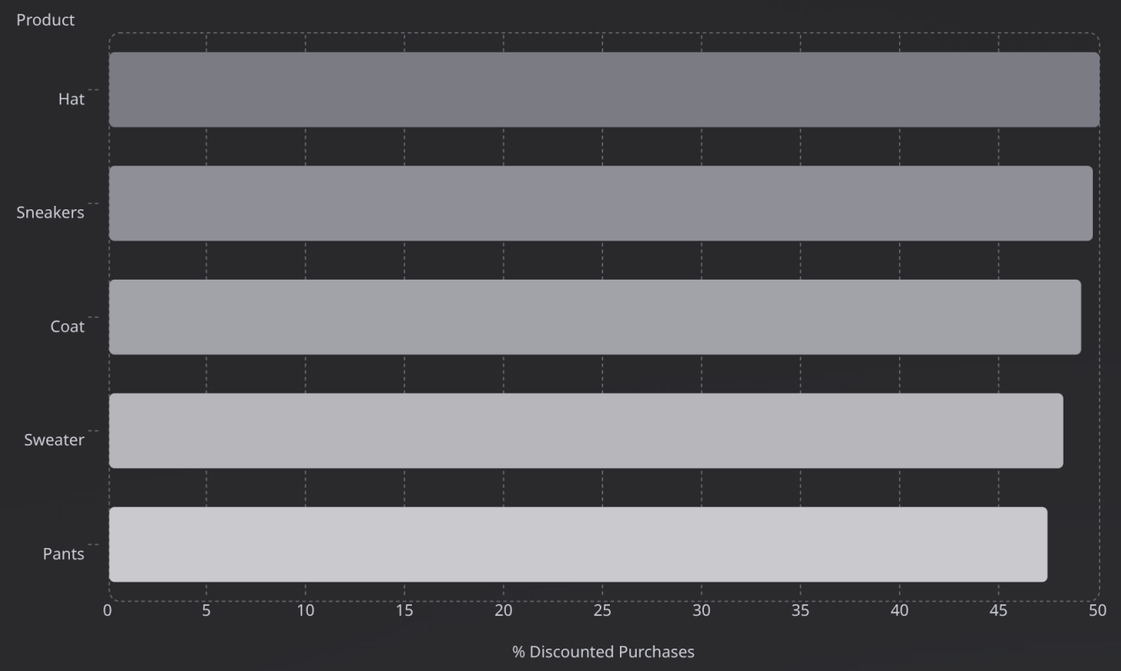
Discount-Dependent Products



Key Finding

~50% of purchases for these 5 products involve a discount — review margin impact.

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Customer Segmentation

Loyal

3,116 customers

Returning

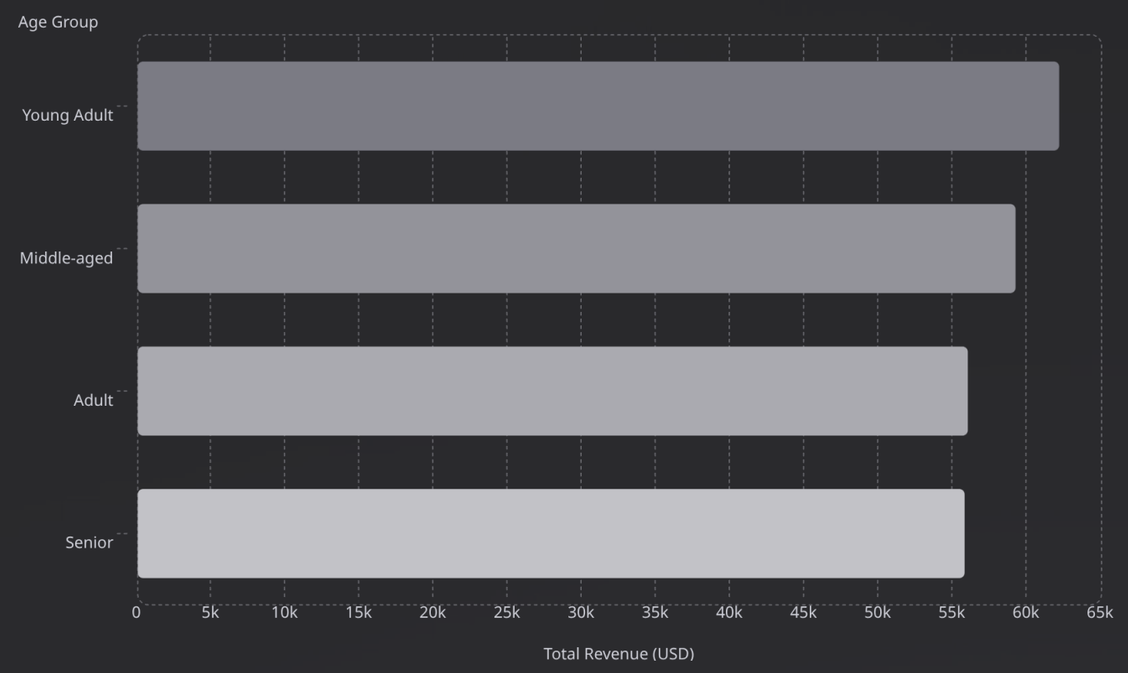
701 customers

New

83 customers

Of 3,476 repeat buyers (5+ purchases), **958 are subscribers** — repeat buyers are more likely to subscribe.

Revenue by Age Group



Insight

Revenue is **evenly distributed** across age groups — Young Adults lead slightly at $62K.

Top 3 products per category: Blouse, Pants, Shirt (Clothing); Sandals, Shoes, Sneakers (Footwear); Jewelry, Sunglasses, Belt (Accessories); Jacket, Coat (Outerwear).

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers → Loyal segment

Review Discount Policy

Balance sales lift with margin control

Targeted Marketing

Focus on high-revenue age groups & express shippers

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